The Haynes-Shockley experiment

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January 2022

Abstract

Through the Haynes-Shockley setup, transport properties of holes in an n-doped germanium sample are investigated. Classical diffusion and transport theory are able to estimate mobility, diffusion constant and recombination time. Slight differences with literature values are supposed to be caused by systematic experimental errors and by quantum effects.

Introduction

The Haynes-Shockley [2] experiment purpose is to measure mobility, diffusion constant and recombination lifetime of minority carriers inside a doped semiconductor. A beam of photons generates an excess concentration of minority carrier in one side of the bar-shaped sample. The minority carriers are forced to move by an applied sweep potential and collected after a certain time on the opposite side of the sample. Knowing the distance d between the target of the beam and the collector, the drift velocity of carriers v_d may be measured. If a n-doped semiconductor is taken in exam, mobility of holes can be expressed as $\mu_p = v_d/E_S$, where $E_S = V_S/L$ is the sweep field and L is the length of the sample. Indeed, only ambipolar mobility μ can be measured; however, in a heavily n-doped semiconductor

$$\mu = \frac{\mu_n \mu_p (n-p)}{p\mu_p + n\mu_n} \xrightarrow[p\to 0]{} \mu_p.$$

According to diffusion theory, an excess of carriers represented by a δ function centered at $x = 0$ at $t = 0$ will spread following a gaussian distribution. The peak will move under the applied field E_S and will decrease in intensity due to recombination processes. If minority carriers are characterized by a recombination time τ, the excess carrier density with respect to equilibrium value is

$$\Delta p(x,t) = \frac{\Delta p(0,0)}{2\sqrt{\pi D t}} e^{-t/\tau} \exp\left(-\frac{(x-\mu_p E_S t)^2}{4Dt}\right),$$

where D is the diffusion coefficient. If the contact between collector and sample is rectifying, the current intensity – and then the voltage drop across an internal resistance – is proportional to minority carrier density.

Materials and Methods

The sample is an n-doped germanium bar with dimensions $3 \times 3 \times 30$ mm, the sweep voltage is applied at the ends, in which two ohmic contacts are realized via soft soldering. Theese contacts ensure that the field inside the sample is proportional to V_S. The (partially) rectifying contact at the collector is made by a needle touching the bar, realizing a metal-semiconductor contact. Photons with $\lambda = (905 \pm 10)$ nm are generated by a laser diode with energy $\hbar\omega$ greater than the gap energy E_g and carried at the sample surface with an optic fiber. The experimen-

tal apparatus allows to vary: the distance d between needle and optic fiber end; the amplitude of sweeping voltage square wave V_S; the amplitude of the voltage pulse V_L applied to laser diode; the delay t_{delay} between the rising edge of the sweeping wave and laser pulse. Moreover, a differential amplifier processes the collector output V_c, partially removing the sweeping voltage. From the SHIFTED OUT channel the output is $V_c - \alpha V_S$, where α is a controllable parameter.

Data Acquisition

Voltage measures are achieved with a digital oscilloscope, with a resolution of 8-bit ($\simeq$ 250 ADC channels) and 10 divisions. Every division has 25 channels, therefore the sensitivity is 4% of the Volt/div value. For the Voltage vs. time measurements the semi-amplitude of the sensitivity is chosen as a fixed y-axis error, which corresponds with a 2×10^{-4} V error. Varying the differential amplifier parameter α and the trigger level of the oscilloscope, the gaussian shape in the collector output can be displayed. In order to remove the noise, bringing the signal far from the rising edge of V_S, t_{delay} is increased. The laser pulse is followed by an electromagnetic signal that cross the bar with (almost) instat speed, always preceding the gaussian signal and fixing the $t = 0$ condition on the timescale. Once the signal is detected and isolated from the noise, an averaging between multimple acquisitions is made automatically by the oscilloscope.

Results

I-V characteristics of the ohmic and the metal-semiconductor contacts have been measured in order to verify the expected behaviour (Fig. (1)). The parameters of interest for this experiment are V_S, d and V_L. Fig. (2) displays common features of all data: laser's signal is always fixed at $t = 0$ and for some $t > 0$ the gaussian signal emerges from the background noise. In Fig. (3) the gaussian fit functions of some part of the data are displayed, separated from the noise. In every figure only one parameter is varied.

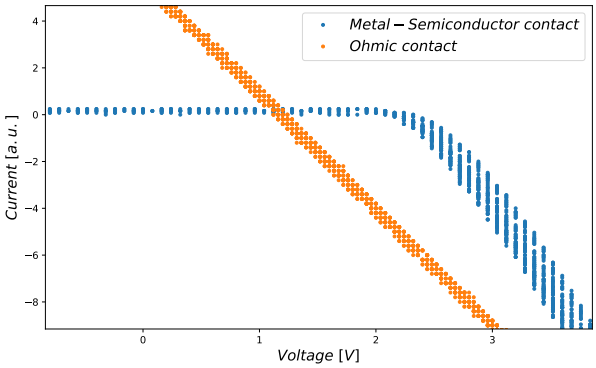


Figure 1: I-V characteristics of ohmic contacts at the ends of the sample and metal-semiconductor contact. From the figures it is clear that linear (orange) and Schottky (blue) behaviour is respected.

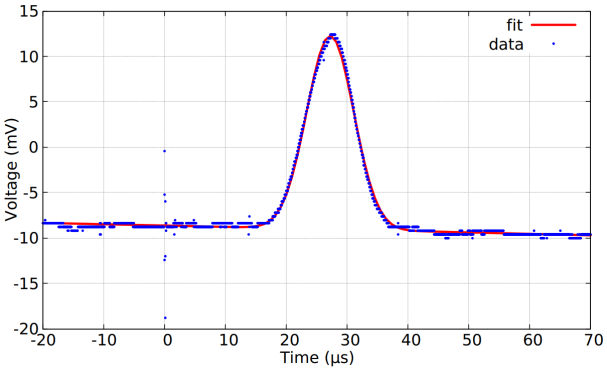
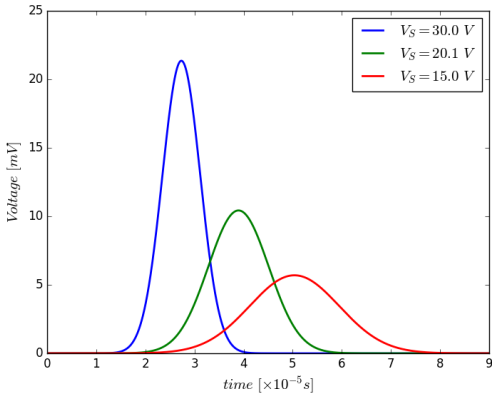
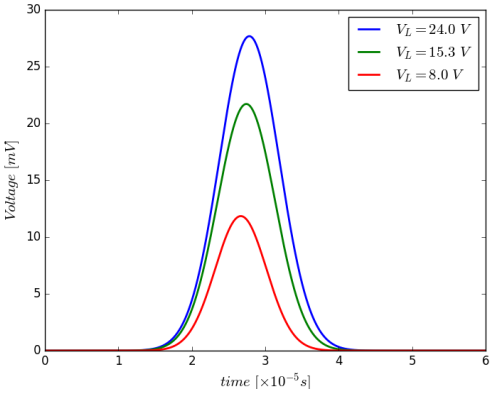


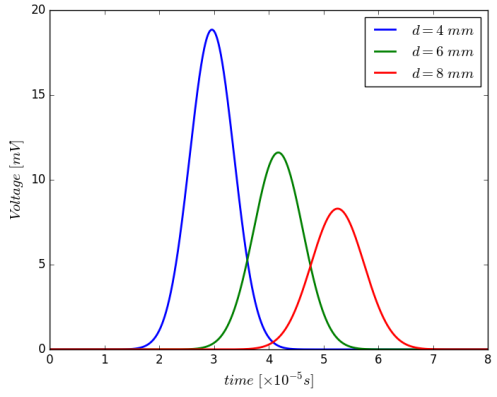
Figure 2: An example of data acquisition. The few points far from background at $t = 0$ are the laser's signal, fit is achieved in the range $t > 0$, the function is a sum of an exponential background and a gaussian.



(a)



(b)



(c)

Figure 3: Gaussian fit functions of the collector's signal V_C. In (a) V_S is varied, in (b) V_L is varied, in (c) d is varied. Only one of the three parameter at a time is varied.

Discussion

From Fig. (1) it is clear that the end contacts have a linear ohmic behaviour. The I-V characteristic of collector contact determines the doping type of the sample, in this setup the germanium bar is n-doped. This step is needed to choose the correct direction of E_S across the sample.

The sweeping signal is a square wave, therefore in absence of carrier injection collector's signal stabilizes exponentially after a certain commutation time. For this reason, injection must occur with a delay bigger than the comutation time. Furthermore, the best guess for background signal is an exponential. The time of flight t_f is expected to depend on V_S and d. With respect to Fig. (3), it is clear that increasing only V_L, the peak position on the timescale remains fixed, while the height of the peak increase. This is due to conservation of charge: if the recombination process is not taken into account, the area of the collector's signal is conserved. In fact, diffusion only increase the width of the gaussian in time and E_S only move rigidly the gaussian with velocity v_d. Integrating the full expression of Δp,

$$A(t) = \int_{-\infty}^{\infty} \Delta p(x,t)dx = \Delta p(0,0)e^{-t/\tau}.$$

The integral of collector signal is proportional to $A(t)$ and therefore it shares the same functional form. Measures of the signal when t_f is too high are difficult to perform, because the signal will spread out until it is no more possible to separate it from noise.

Collector's voltage V_C, proportional to the minority charge excess Δp, is measured with respect to time and with $x = d$ fixed. Furthermore, from Fig. (2) it is clear that the fraction of recombined carriers is negligible in the time interval in which the excess carriers pass over the collector's needle, because the signal is symmetric and gaussian-like. For the same reason, the time-varying standard deviation is taken constant. With theese assumptions, the fit parameters, calculated in $t = t_f$, are:

- the amplitude $a \sim \dfrac{\Delta p(0,0)e^{-t_f/\tau}}{\mu E_S}$;

- the standard deviation $\sigma = \dfrac{\sqrt{2Dt_f}}{\mu E_S}$;
- the mean, equal to t_f.

From the fit in Fig. (4) the recombination time τ can be estimated as the angular coefficient of $\ln(A) = \text{const} + \ln(a\mu E_S)$.

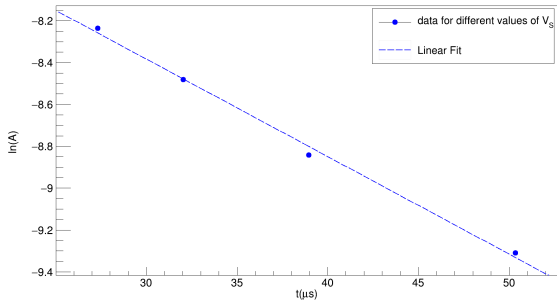


Figure 4: Linear fit of $\ln A$ *as a function of time of flight* t_f*. Different* t_f *are obtained varying* V_S

The mobility can be directly estimated for every data with the formula $\mu = (Ld)/(t_f V_S)$. Fig. (5) collects all the values of μ: the mobility is approximately constant, the best estimation is the mean of theese values.

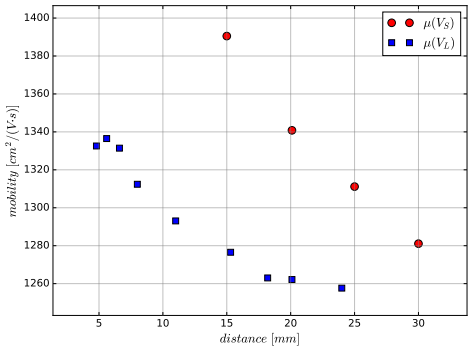
The diffusion constant can be estimated from the standard deviation, performing a linear fit (Fig. (6)).

Conclusions

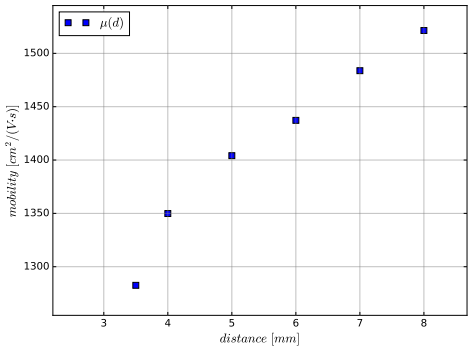
The table below summarizes the results and offers accurate values of the analyzed quantities from literature. [1]

	$\mu[cm^2/Vs]$	$D[cm^2/s]$	$\tau[\mu s]$
experimental	1340 ± 70	53 ± 4	21 ± 1
literature	1900	50	20

While diffusion constant is comparable with the exact value, all the values of mobility in Fig. (5) are far below the exact value of germanium hole's mobility. This effect is possibly due to a systematic error in the measure of the voltage difference between the



(a)



(b)

Figure 5: Results of hole's mobility as a function of the varying parameter. In (a) V_S *and* V_L *is varied, in (b)* d *is varied. Mobility should be approximately constant.*

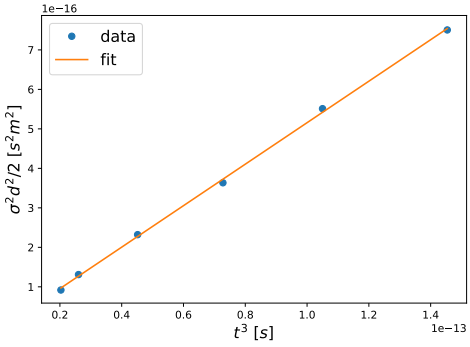


Figure 6: Linear fit of diffusion constant D*.*

ends of the sample V: if r is the resistance of the ohmic contacts and R is the resistance of the germanium bar, $V = V_S/(1 + 2r/R)$. While diffusion constant is estimated with only d and t_f parameters ($v = \mu E_S = d/t_f$), mobility strongly depends on $E = V/L = V_S/(1 + 2r/R)L$. We can define an effective length $L_{eff} = (1 + 2r/R)L$, which is different from the original length of the sample if ohmic contacts resistance cannot be neglected. Further, carrier injection takes place near the surface, therefore crystal periodicity is not respected and holes can scatter more easily. Of course, Einstein's relation is not respected: $D/\mu = k_B T/e \simeq 26$ mV, while the estimated value is (40 ± 5) mV. There is also a systematic error when measuring μ varying V_S, which applies for small V_S. Indeed, even if $V_S = 0$, the collector measures an increasing and decreasing excess carrier, due to diffusion and recombination. In this dissertation we avoided this problem simply taking measures of mobility for high sweeping voltages, for which this effect is negligible.

References

[1] *Electrical properties of Germanium*. URL: http://www.ioffe.ru/SVA/NSM/Semicond/Ge/electric.html.

[2] A. Sconza, G. Galet, and G. Torzo. "An improved version of the Haynes–Shockley experiment with electrical or optical injection of the excess carriers". In: *American Journal of Physics* 68.1 (2000), pp. 80–87. URL: https://doi.org/10.1119/1.19376.